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Inhibition of mild steel corrosion in 0.5 M HCl medium using eggshell extract: experimental, electrochemical and optimization studies

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Abstract

Corrosion is one of the major challenges in industries. This study used Box Behnken Design (BBD) to examine the effectiveness of eggshell extract as green inhibitor. Physicochemical analysis was done to determine the major physicochemical constituents. The process variables optimised study were: eggshell extract (0.2–0.8 g/L), immersion time (3–12 days) and temperature (30–60 °C). Characterization was done using Fourier Transform Infrared Spectroscopy (FT-IR); Scanning Electron microscope (SEM). Polarization study was used to determine type of inhibitor. The best process level was 7.5, 0.8 g/L, 30 °C, with corrosion rate of 0.00000049 mm/yr; while 0.00000219 mm/yr at: 9 days, 0.8 g/L at 35.5 °C was observed at optimal process level (validated). The FTIR and SEM showed that mild steel from validated experiment had higher adsorption, with more protective layer formed due to presence of more heteroatoms. Polarization study showed that the extract was a mixed-type inhibitor. The results of physicochemical, Polarization, FTIR and SEM analyses supported the findings that eggshell extract can reduce corrosion rate by adsorption on the metal surface.

Keywords Eggshell extract, Corrosion, Inhibition efficiency, Polarization, Optimization, Weight loss

1 Introduction

Corrosion is a natural process that occurs when metals react with environment. This reaction causes them to deteriorate over time. It happens due to chemical reactions between the metal and substances like oxygen, water, acids, or salts [1]. Corrosion is a major problem frequently encountered in the oil and gas pipelines, refineries and petroleum industries [2]. Therefore, it is important to use corrosion techniques to retard the effects of metal dissolution. The majority of conventional inhibitors are organic compounds that are usually non-biodegradable and toxic to the environment. Green corrosion inhibitors being an effective and environmentally friendly technique have attracted more attention in recent years [3, 4]. Green inhibitors are extensively used to control



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the corrosion of mild steel, aluminum and carbon steels in acidic environments, especially in the oil and gas industry [5]. To address the high expense of corrosion inhibitors, researchers are focusing on green corrosion inhibitors that are biodegradable and non-toxic [6]. Many efforts shown the role of different plant extracts in the protection of mild steel surfaces in different corrosive environments. Some researches already reported on plant extracts as inhibitors are: Eucalyptus leaf [7]; *Corchorus olitorus* stem [8]. Moreover; Proteins and Amino Acids had been studied as green inhibitors (biowastes) that adsorbed on metal surfaces also provide protective layer that lowers rate of corrosion [8, 9]. Extracts from algae and certain microbes contained bioactive compounds that also inhibit corrosion through adsorption and film formation [10]. Previous research has been done on the use of eggshell extract as a corrosion inhibitor in 30wt % wt NaCl [11]. Previous study on Box Behnken Design (BBD) as experimental design for corrosion studies are [12–15]

The gap for corrosion inhibitors is justified by the reasons that corrosion causes colossal economic and industrial challenges and losses. As a fundamental progressive step in this quest, it has been found that the use of corrosion inhibitors is one of the most practicable ways of protecting metals in different corrosive environments.

There is sparse research on eggshell extract optimisation study as an inhibitor in HCl acidic environment. This study aims to investigate the effectiveness of eggshell extract as corrosion inhibitor in 0.5 M HCl using experimental, electrochemical and optimization studies.

2 Experimental

The materials used in this research were Eggshell which is readily available and HCl (for the corrosive environment). Weight loss measurement using an optimization approach (Box-Behnken Design).

2.1 Experimental techniques

2.1.1 Preparation of eggshell extract

Eggshells were gathered from the Landmark University campus. Mild steel was properly cleaned with distilled water, air dried, and milled.

Maceration method was used for the extraction process.

2.1.1.1 The procedure The eggshells were first washed. After that, it was dried and processed into a fine powder (to increase the surface area) and facilitates better ethanol penetration. A closed container with a stopper, was filled with the 250 g of grinded eggshell. It was thereafter covered with 500 millilitres of ethanol. Eggshells and ethanol were used in a 1:2 (w/v) solvent ratio. Filtration was used to separate the liquid extract from the remaining solid particles after three days of soaking in the sealed container with sporadic stirring.

2.1.2 Preparation of corrosive medium

Distilled water was prepared with 48.3 ml of HCl to make 1000 ml of 0.5 M HCl. The reagent used was analytical grade with a CAS number of 7647-14-5.

2.1.3 Mild steel preparation

The mild steel of dimensions 2.2 cm X 1.9 cm X 2 cm was used. Hole of 0.3 cm was drilled in the middle for total immersion in the environments. Mild steel was degreased with ethanol, cleaned with distilled water, and then dried with acetone.

2.1.4 The physicochemical analysis

The eggshell's components were determined using physicochemical analysis. The components analysed for were: Protein and nitrogen, Fat and Moisture content.

Protein and nitrogen analyses were performed using the usual Kjeldahl method [16, 17].

Nitrogen: The sample (0.5 g) was digested using strong sulphuric acid and catalysts (such as copper sulphate and sodium sulphate) to transform organic nitrogen into ammonium sulphate. Distillation: The digest is neutralised with sodium hydroxide to release ammonia, which is subsequently distilled into a boric acid solution.

protein: The percentage of crude protein is obtained by multiplying the percentage of nitrogen by a protein conversion factor, which is commonly 6.25.

Fat: To extract lipid content from dried eggshell powder for fat analysis, Soxhlet extraction was performed using a nonpolar solvent, usually petroleum ether or ethyl ether.

Moisture content was measured by oven drying (often at 105 °C). Moisture was determined through a gravimetric oven-drying method (AOAC 925.09).

2.2 Weight loss measurement

Table 1 shows Ranges and Variables while Table 2 is for matrix interaction generated from the BBD. Table 3 is for 17 experimental runs generated from Box-Behnken Design (BBD). Software used was Version 13 of Design Expert. The 17 experimental runs generated by BBD was used to determine weight loss and corrosion rate, respectively.

The mild steel samples were weighed, then submerged in test tubes containing (0.2–0.8 g/l) solution with time variation of 3–12 days in a water bath at temperature ranges of 30–60 °C. After immersion time of each experiment, the samples were removed, and the weight loss was calculated. The difference between the final weight and the initial weight after immersion was later recorded as the weight loss. Additionally, the corrosion rates (g/cm²days) in the presence and absence of inhibitors was also calculated. The methodology for the weight loss measurement was adapted from [12, 13, 15, 18, 19]. Weight loss was and corrosion rate will be calculated using Eqs. 1 and 2.

$$\Delta W = W_b - W_a \quad (1)$$

where W_b is weight before immersion and W_a is weight after immersion.

The corrosion rate will be calculated using;

$$CR = 87.6 \times \frac{\Delta W}{A_t} \quad (2)$$

Table 1 Ranges and variables

Name	Ranges			
	Units	Low	High	Symbols
Temperature	°C	30	60	A
Time	Days	3	12	B
Inh.Conc	g/L	0.2	0.8	C

Table 2 Matrix interaction generated by the BBD

S/n	Temp	Time	Inh.Conc
1	0	−1	1
2	−1	−1	0
3	0	0	0
4	−1	0	−1
5	0	1	−1
6	0	0	0
7	0	−1	−1
8	1	1	0
9	1	0	−1
10	0	1	1
11	−1	0	1
12	1	−1	0
13	1	0	1
14	0	0	0
15	−1	1	0
16	0	0	0
17	0	0	0

Table 3 Variables interaction for 17 runs generated

Experiment number	Temperature (°C)	Time (days)	Inhibitor Conc (g/L)
1	45	12	0.2
2	45	12	0.8
3	45	7.5	0.5
4	45	3	0.8
5	45	3	0.2
6	60	12	0.5
7	60	7.5	0.2
8	60	3	0.5
9	30	12	0.5
10	30	7.5	0.8
11	30	7.5	0.2
12	30	3	0.5
13	45	3	0.2
14	60	12	0.5
15	60	12	0.5
16	60	12	0.5
17	45	3	0.2

where CR is the corrosion rate, A is specimen area, t is exposure time and ΔW is the weight loss adhering to exposure time.

2.3 Characterization

2.3.1 Fourier transform infrared spectroscopy (FT-IR) Analysis

The Perkin-Elmer-1600 FT-IR Spectrometer model was used to determine active functional groups of mild steel corrosive environments for: optimal process level (verified), and lowest corrosion rate (best level seen from Box Behnken design) and blank.

2.3.2 Scanning electron microscope (SEM)

SEM was used to determine the surface morphology of mild steel under three different conditions: lowest corrosion rate (best), no inhibitor (blank) and optimal process level (validated).

2.4 Polarization study

The mild steel specimen (active area) was mechanically polished using a sequential series of silicon carbide (SiC) abrasive papers. The grinding sequence progressed through 600, 800, 1000, and 1200 grits. To ensure surface uniformity and prevent overheating, polishing was performed under a continuous stream of distilled water. Following the final grit, the mild steel was degreased with acetone, rinsed with deionized water, and dried in a stream of warm air immediately before immersion. Mild steel specimen encased in Versocit acrylic mounts. Counter Electrode (CE) with A high-surface-area platinum (Pt) foil/rod was used. Silver/Silver Chloride of $A = 1.65\text{cm}^2$. As reference Electrode. Potentiodynamic polarization curves were recorded using an Autolab PGSTAT 30 Eco Chimie potentiostat. After reaching a stable Open Circuit Potential (OCP) for 30 minutes, the potential was scanner from -250mV to $+250\text{mV}$ relative to 30 E_{ocp} .

To ensure quasi-steady-state conditions during the anodic and cathodic transitions, a constant scan rate of 1 mV/s was used. The data were recorded at a rate of 10 points per decade. The corrosion potential (V) and current density were estimated using the Tafel plots. The methodology was adapted from the work of [13].

3 Results and discussion

3.1 Result of physicochemical analysis

The result of physicochemical analysis of eggshell extract are shown in Table 4. It showed that protein is present in high amounts, nitrogen, protein, fat and moisture content. where amino acid in the protein is considered as good inhibitor [15]. Since amino acids will be adsorbed onto metallic surfaces to form protective, film-like barrier against corrosive chemicals, its existence would make the extract a good and effective inhibitor. In addition to acting as an active center for adsorption onto the metallic surface, nitrogen, along with other eggshell constituents, enables the inhibitor to achieve high inhibition efficiency. The fat component in eggshell powder help to form a protective film on metal surfaces, functioning as a natural, environmentally friendly corrosion inhibitor. This validated the reports [11, 15, 20].

3.2 Result of weight measurement

Table 5 displayed result of weight loss measurement and corrosion rate of 17 experimental runs as observed. It was observed that the best process level (lowest weight loss and lowest corrosion rate :0.0000049 mm/yr) observed was experiment 10, with process variables of 0.8 g/L; $30\text{ }^{\circ}\text{C}$ and 7.5 days.

Table 4 Result of physicochemical analysis

Bioactive constituents	Amount
Nitrogen	0.92%
Protein	5.79%
Fat	1.5%
Moisture content	0.5%

Table 5 Result of weight loss measurements

Experiment number	Temperature (°C)	Time (days)	Inhibitor Conc (g/l)	W _a (g)	W _b (g)	ΔW (g)	Corrosion rate (mm/yr)
1	45	12	0.2	5.27	5.19	0.09	0.0000179
2	45	12	0.8	5.58	5.46	0.12	0.0000239
3	45	7.5	0.5	5.31	5.22	0.09	0.0000287
4	45	3	0.8	5.49	5.43	0.06	0.0000478
5	45	3	0.2	5.39	5.14	0.25	0.0001994
6	60	12	0.5	5.66	5.26	0.4	0.0000797
7	60	7.5	0.2	5.17	4.86	0.31	0.0000988
8	60	3	0.5	5.30	5.24	0.06	0.0000478
9	30	12	0.5	5.39	5.18	0.21	0.0000418
10	30	7.5	0.8	5.34	5.31	0.03	0.00000490
11	30	7.5	0.2	5.18	4.99	0.19	0.0000606
12	30	3	0.5	5.45	5.40	0.05	0.00003987
13	45	3	0.2	5.39	5.14	0.25	0.0001994
14	60	12	0.5	5.66	5.26	0.4	0.0000797
15	60	12	0.5	5.66	5.26	0.4	0.0000797
16	60	12	0.5	5.66	5.26	0.4	0.0000797
17	45	3	0.2	5.39	5.14	0.25	0.0001994

Table 6 ANOVA

Source	Sum of Squares	df	Mean Square	F- value	P value	
Model	0.1819	3	0.0606	6.70	0.0057	Significant
A (Temp)	0.0682	1	0.0682	7.53	0.0167	Significant
B (Time)	0.0423	1	0.0423	4.68	0.0498	Significant
C (Inh.Conc)	0.0371	1	0.0371	4.10	0.0641	
A ²	0.012	1	0.012	1.54	0.2546	
B ²	2.21	1	2.21	0.28	0.6120	
C ²	9.89	1	9.89	0.13	0.7339	
AB	0.019	1	0.019	2.43	0.1627	
AC	5.28	1	5.28	0.067	0.829	
BC	0.024	1	0.024	3.03	0.1255	
Residual	0.1176	13	0.0090			
Lack of fit	0.1176	8	0.0147			
Pure Error	0.0000	5	0.0000			
Cor Total	0.2995	16				
R ²	0.9935					
Adjusted R ²	0.9851					

3.3 ANOVA

The ANOVA is reported in Table 6. Model with p-value of.

0.0500 suggested that model is significant., A and B as significant terms. The values of R² (0.9935) and Adj R² (0.9851) showed high level of correlation.

The regression equation in actual terms is as stated in Eq. 3.

$$\begin{aligned} \text{Corrosion rate} = & +1.74533\text{E} - 004\text{temp.} + 7.10973\text{E} - 007\text{Xtime} - 1.15587\text{E} - 004\text{XInh.Conc} \\ & + 2.09898\text{E} - 007 \text{ temp X time} - 2.98315\text{E} - 006 \text{ X tempX inh. Conc} \\ & + 2.28501\text{E} - 005 \text{ X time X inh. Conc} \end{aligned} \quad (3)$$

3.4 3D surface plots

The 3D plots are as shown in Figs. 1, 2 and 3 respectively.

Factor Coding: Actual

3D Surface

CorrosionRate (mm/yr)

Design Points:

● Above Surface

○ Below Surface

4.9E-06 0.0001994

X1 = B

X2 = C

Actual Factor

A = 45

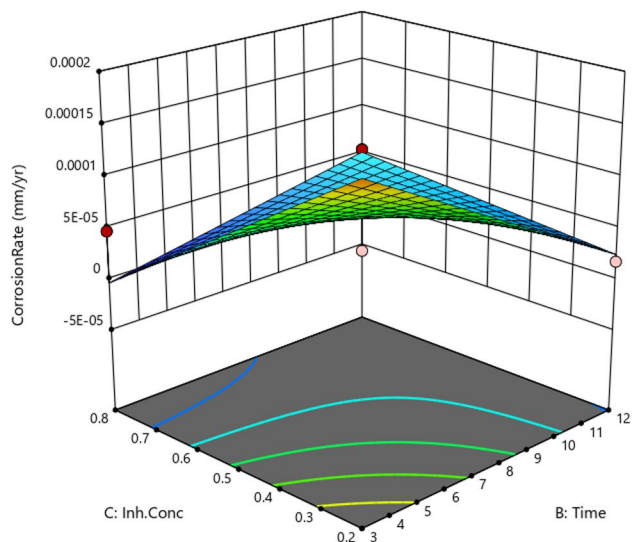


Fig. 1 Plot of CR vs. inh.con and time

Factor Coding: Actual

3D Surface

CorrosionRate (mm/yr)

4.9E-06 0.0001994

X1 = A

X2 = B

Actual Factor

C = 0.276321

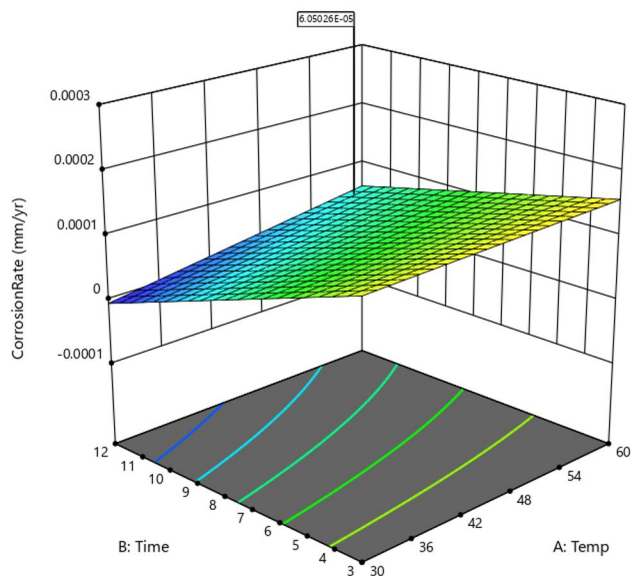


Fig. 2 Plot of CR vs. temperature and time

Figure 1 shows that as the immersion time and inhibitor concentration increase, the corrosion rate decreases. Figure 2 shows that an increase in temperature and time increases the corrosion rate. Figure 3 demonstrates that as the Inhibitor conc increased with an increase in temperature, the corrosion rate decreased. This supported the findings of [8, 15, 20].

Factor Coding: Actual

3D Surface

CorrosionRate (mm/yr)

● Design Points

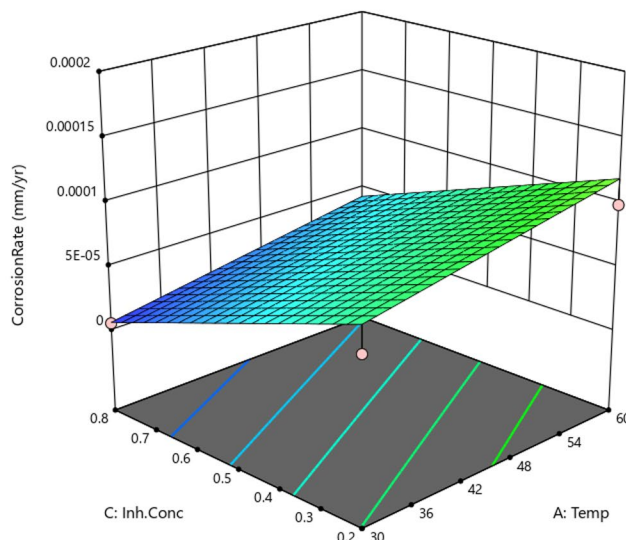
4.9E-06 0.0001994

X1 = A

X2 = C

Actual Factor

B = 7.5

**Fig. 3** Plot of CR vs. Inh. conc. and temperature

3.5 Validated experiment result

The optimal process level generated are variables (temp: 35.5 °C, time of immersion: 9 days and Inh. Conc: 0.63 g/l which was generated by the BBD and it was validated. The corrosion rate observed from the experimental validation of the optimal process level was 0.000000219 mm/yr. This result is lower than what was reported by [18, 19].

3.6 Results of FTIR (Fourier-Transform Infrared Spectroscopy) analysis

The FTIR results shown in Figs. 4, 5, 6 are corrosive environments of: Blank lowest corrosion rate and validated optimal process level. The peaks at: 3298.76, 3197.24, 2798.46, 1550.32, 1274.38, 1200.96, 897.62 and 515.22 are linked to O-H, -CH₂, C=N, C=C, -NO₂, CH₃, -C-NH₃, C-H, for the blank mild steel corrosive environment as shown in Fig. 7. the peaks at 3298.75, 3197.42, 2788.36, 1549.53, 1274.41, 1200.85, 898.42, and 512.22 are O-H, -CH₂, C=N, C=C, -NO₂, CH₃, -C-NH₃, C-H, for the mild steel of lowest corrosion rate environment (best level process) with absorption of the extract molecules on the mild steel of the corroded products as shown in Fig. 5. The peaks 3212.73, 3148.02, 2711.98, 1550.30, 1274.38, 1200.87, 898.21 and 515.22 as shown in Fig. 6 are: O-H, -CH₂, C=N, C=C, -NO₂, CH₃, -C-NH₃, C-H with more adsorption of the extract during the immersion that blocked the corrosion in the optimal process variable environment. This finding validated [20].

3.7 Results of SEM studies

The SEM morphology in Fig. 7 revealed pitting corrosion and significant surface degradation. As shown in Fig. 8 passive film was visible on the surface with lowest corrosion rate (best) because eggshell extract was present. Mild steel of optimal process level (validated) had more white spots that indicated sufficient adsorption of corrosion inhibitor and the lessens dissolution rate as shown in Fig. 9 [20, 21].

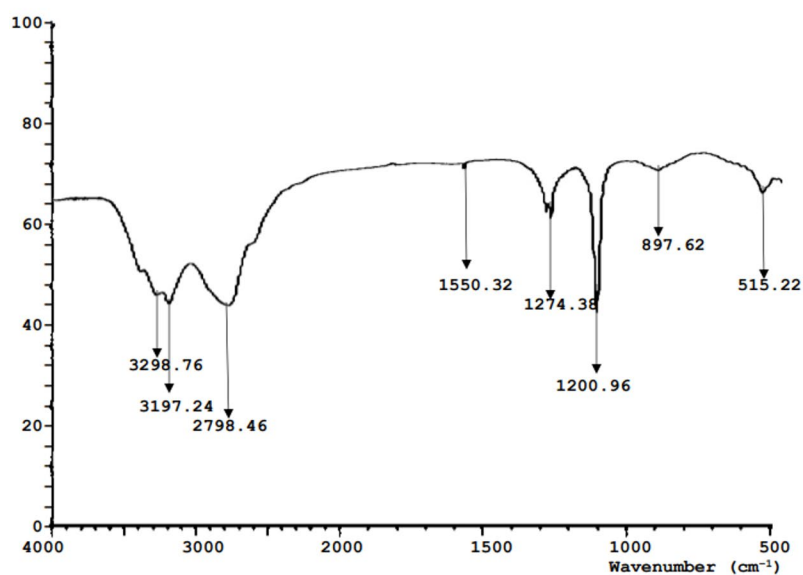


Fig. 4 FT-IR of corrosive environment for blank mild steel

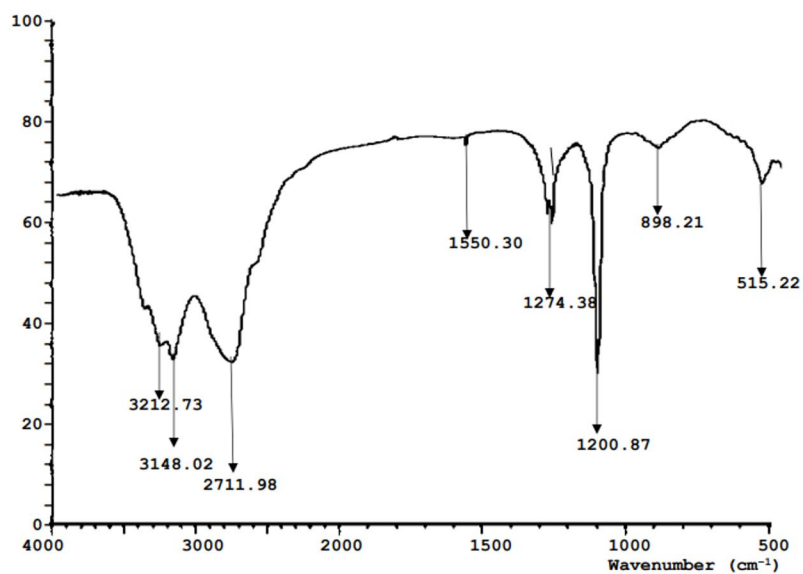


Fig. 5 FT-IR of corrosive environment for best process level

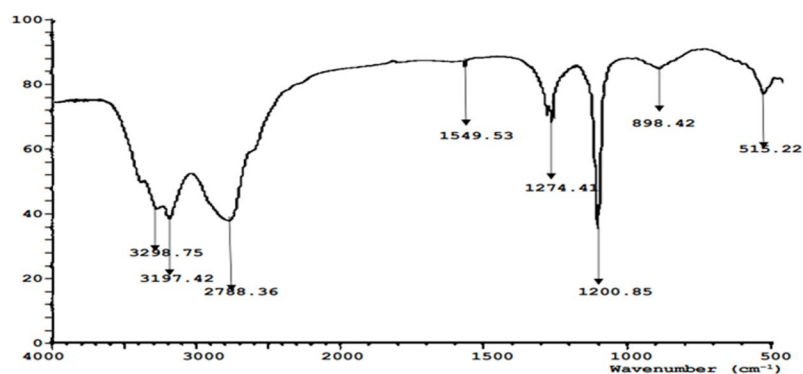


Fig. 6 FTIR of optimal process level (validated) corrosive environment

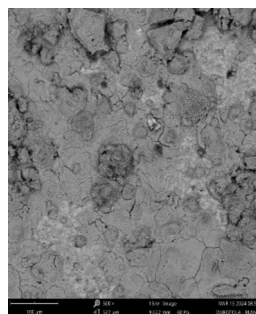


Fig. 7 SEM of blank mild steel

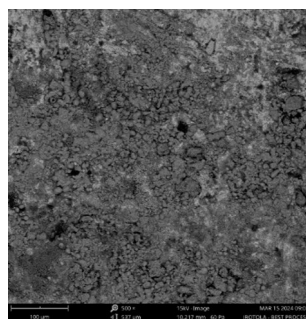


Fig. 8 SEM of mild steel for lowest corrosion rate (best)

3.8 Result of polarization

The result of polarization is in Table 7. It showed that at 0.2 g/L; corrosion rate dropped to 1.583 mm/Yr with IE of 79.66% while at 0.8 g/L the corrosion rate observed was 0.061 mm/yr. with IE of 99.21% respectively. Additionally, it was noted that no discernible effect was seen at greater than 0.8 g/L inhibition concentration.

The result of IE in this study is higher than 94% [22] and the corrosion rate lower than 0.009 mm/yr [23] and 0.31 mm/yr [24]. The polarization resistance (PR) in the inhibited at 0.8 g/L (4863.00 Ω) is higher than PR of the control (38.14 Ω) which suggested that the extract is a good inhibitor [13, 25, 26] Results of Tafel plots and corrosion densities

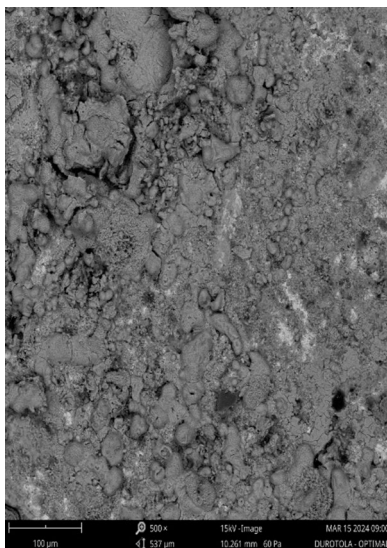


Fig. 9 SEM of mild steel for optimal process level (validated)

further suggested that the extract has a strong anti-corrosion ability that acted as mixed inhibitor which confirmed [13, 27–33]. The polarization plot is as shown in Fig. 10

4 Conclusion

1. The result of physicochemical analysis shows the presence of nitrogen, protein, fat and moisture content. The high amount of protein will form a film that will act as a blockage against corrosion because proteins are long chains of amino acids.

The impacts of the results are:

- Best process level: 30 °C, 7.5 days, and 0.8 g/L extract at experiment 10.
- The optimal process level evaluated resulted in corrosion rate 2.19×10^{-5} mm/yr.
- FT-IR analysis revealed that more adsorption was formed on the mild steel surface of optimal level process (validated experiment) than on mild of best process level.
- The polarisation investigation revealed that the 80% inhibitor concentration (0.8 g/l) had the lowest corrosion rate, an inhibition efficiency of 99.2 1%, and a corrosion current density of 0.00000528 A/cm². The extract was inferred to be a mixed-type inhibitor.
- Result of SEM confirmed that more protective film formed on mild steel of the optimal process level validated and FTIR result of the optimal process level corrosive environment showed more adsorption of inhibitor.
- This study showed that eggshell extract is a good corrosion inhibitor that is ecofriendly and highly efficient in the acidic corrosive environment.

The future challenge for this study will be variation in composition (e.g. calcium carbonate, protein). This will depend on the source of the eggshell

Table 7 Result of polarization

Extract conc (%)	MS Corrosion Rate X10 ⁻⁵ (mm/y)	Extract inhibition efficiency (%)	Corrosion current (A)	Corrosion current density(A/cm ²)	Corrosion potential(V)	Polarization resis- tance RP(Ω)	Cathodic poten- tial (BC)	Anodic po- ten- tial (Ba)
Control	7.781	-	6.69E-04	6.69E-04	-0.447	38.41	-5.809	10.650
0.2	1.583	79.66	1.36E-04	1.36E-04	-0.421	188.80	-4.843	15.760
0.4	0.711	90.87	6.11E-04	6.11E-04	-0.448	420.50	-5.22	13.710
0.6	0.554	92.87	4.76E-4	4.76E-04	-0.486	539.40	-7.247	14.320
0.8	0.061	99.21	5.28E-04	5.28E-04	-0.716	4863.00	-4.486	5.802
1	0.059	99.24	5.08E-04	5.08E-04	-0.706	4973.00	-4.292	2.289

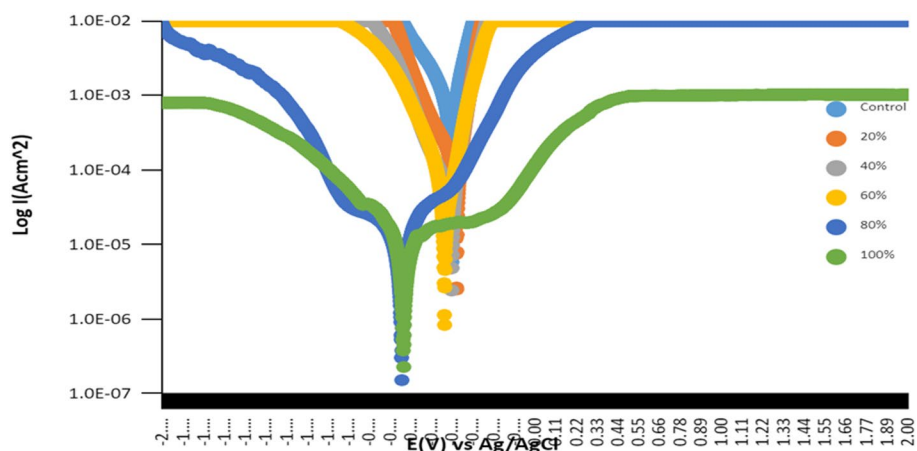


Fig. 10 Result of polarization plot

Acknowledgements

All authors appreciate the entire Landmark University for providing the conducive environment for the research.

Author contributions

O.O: Conceptualization, Data curation, Methodology, Project administration, Software, Supervision, Validation, Writing of original draft and editing; J.B.A: Methodology, Software, Validation, Writing of original draft and editing; A.A: Methodology, Software, Validation; G.O.L: Data curation, Methodology, Resources, Software, Validation; Z.H-H.: Data curation, Methodology, Resources, Software, Validation; S.T.A.: Software, Validation.

Data availability

All data supporting the findings of this study are available within the paper.

Declarations

Ethics approval and consent to participate

This study did not involve human participants or animals. Therefore, ethical approval was not required.

Consent for publication

All authors agree to publish. The authors declare no conflict of interest.

Competing interests

The authors declare no competing interests.

Received: 6 February 2026 / Accepted: 27 June 2026

Published online: 20 July 2026

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